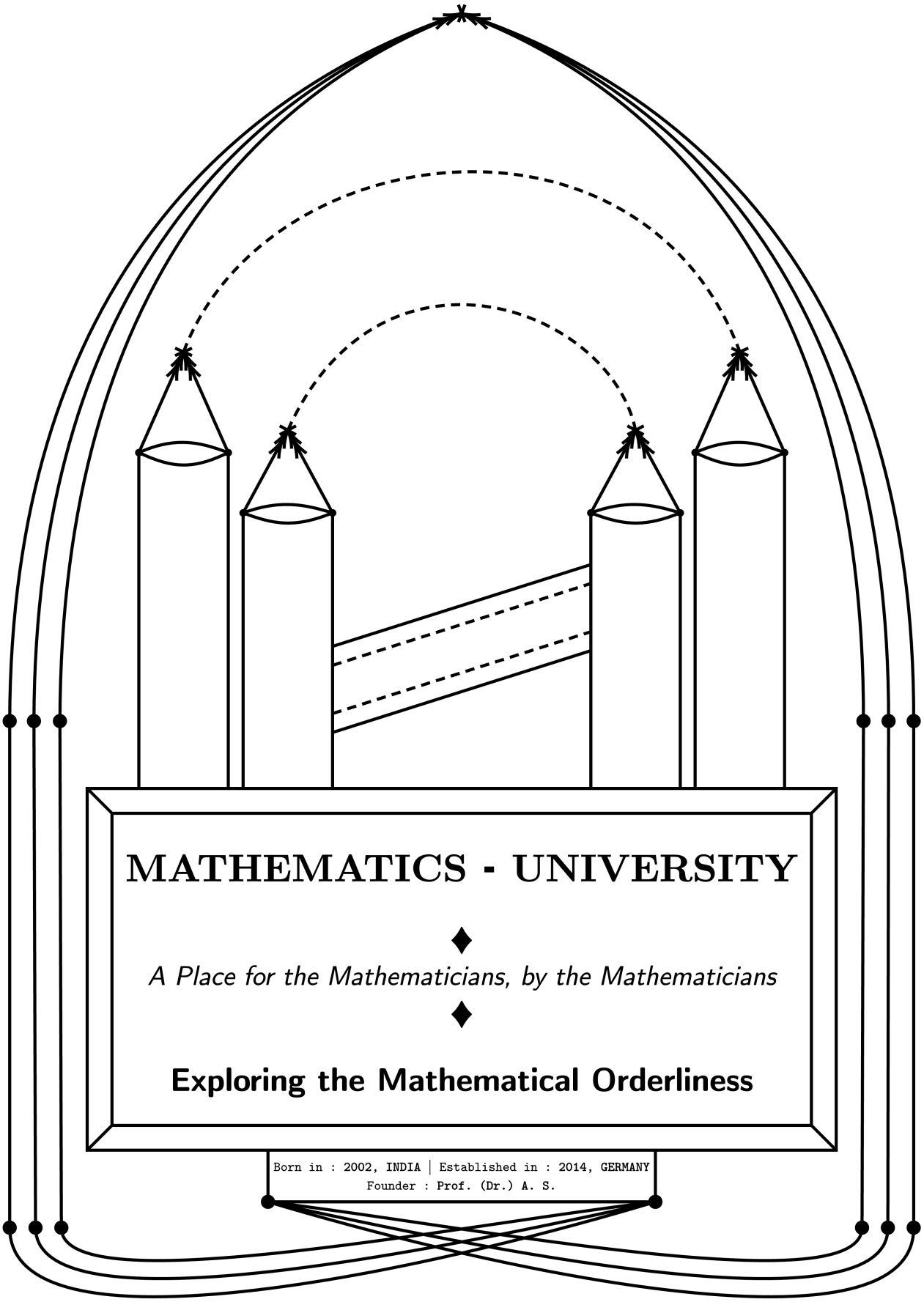

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Sample Question Paper for 2026

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MATHEMATICS - UNIVERSITY

Sample Question Paper for **M.U$^{\infty}$.E.E.**, **2026**

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- MU releases a sample paper for **M.U^{0}.E.E.** and **M.U$^{\infty}$.E.E.** every year. This particular sample paper is for the exam **M.U$^{\infty}$.E.E.**.
- Students must keep in mind that this does not imply to be the exact matches for the original question paper. This is just a glimpse of what you are going to face in the examination hall.
- MU has a vast syllabus for both the exams, though we do not cover each major topics in the sample paper, we take a few topics randomly and make questions accordingly. Original question paper will always contain questions from each and every topic which is there in the official syllabus, so we do recommend you studying whole syllabus by giving your best.
- This sample paper contains a total of 10 questions, 50 marks for each i.e. a total of 500 Marks.

Question in this particular sample paper are made for the following major topics / branches of Mathematics.

- Harmonic Analysis
- Complex Analysis / Geometric Analysis, Linear Algebra
- Functional Analysis + Ergodic Theory
- Algebraic Geometry
- Analytic Number Theory
- Category Theory
- Diophantine Geometry / Diophantine Approximation in projective space.
- Quantum Information Theory / Quantum Thermodynamics inside Mathematical Physics / Quantum Physics

Question 1. For any $p \neq 2$, is the geometric condition

$$\sup_{\theta \in S^1} \int_0^{\delta_0} \Lambda(\theta, \delta)^p \frac{d\delta}{\delta} < \infty$$

sufficient for $\mathcal{M}$ to be bounded on $L^p(\mathbb{R}^2)$?

Question 2. (a) Let S, X_1 and X_2 be Riemann surfaces and let

$$S \xrightarrow{f} X_1 \xrightarrow{g} X_2$$

be orientation preserving diffeomorphisms. Prove that

$$\mu_g \circ f = \left(\frac{\partial f}{\partial z} \Big/ \overline{\left(\frac{\partial f}{\partial z} \right)} \right) \cdot \frac{\mu_{g \circ f} - \mu_f}{1 - \overline{\mu}_f \, \mu_{g \circ f}}$$

(b) Prove the following lemma about compositions of quasiconformal maps: Suppose X, Y and Z are Riemann surfaces and $f : X \to Y$ and $g : Y \to Z$ are orientation preserving diffeomorphisms. Then the following holds:

1. We have that
$$K_f \geq 1$$
with equality if and only if f is a biholomorphism.

2. We have that
$$K_{g \circ f} \leq K_g \cdot K_f$$

3. Finally,
$$K_{f^{-1}} = K_f$$

Question 3. (a) Let H be a Hilbert space and $U : H \to H$ a unitary operator. Let

$$H_{\text{rat}} = \overline{\text{Span}\Big\{ f \in H : \exists i \in \mathbf{N},\ U^i f = f \Big\}},$$

$$H_{\text{tot.erg}} = \left\{ f \in H : \forall i \in \mathbf{N},\ \left\| \frac{1}{\mathbf{N}} \sum_{n=0}^{\mathbf{N}-1} U^{in} f \right\| \to 0 \right\}$$

Show that $H = H_{\text{rat}} \oplus H_{\text{tot.erg}}$

(b) Prove the following version of the van der Corput trick: if $\{v_n\}_{n \in \mathbf{N}}$ is a bounded sequence in H such that for every $h \in \mathbf{N}$, $\lim_{N \to \infty} \frac{1}{N} \sum_{n=1}^{N} \langle v_{n+h}, v_n \rangle = 0$, then

$$\left\| \frac{1}{N} \sum_{n=1}^{N} v_n \right\| \to 0$$

Question 4. Consider the curve

$$\overline{K} = \mathbf{V}(y_1^3 y_2 + y_2^3 y_0 + y_0^3 y_1)$$

in the projective plane, defined over the field $\mathbb{F}_8 = \mathbb{F}_2[\alpha]/\langle \alpha^3 + \alpha + 1 \rangle$.

1. Show that $\overline{K}$ is smooth of degree 4, hence has genus $g = 3$.
2. Show that $\overline{K}$ has 24 points rational over $\mathbb{F}_8$. First show that the three points $Q_0 = (1, 0, 0)$, $Q_1 = (0, 1, 0)$ and $Q_2 = (0, 0, 1)$ are $\mathbb{F}_8$-rational points on $\overline{K}$.
3. Show that the mappings

$$\psi(y_0, y_1, y_2) = (\alpha^4 y_0, \alpha y_1, \alpha^2 y_2) \qquad \tau(y_0, y_1, y_2) = (y_1, y_2, y_0)$$

   take the set $\overline{K}(\mathbb{F}_8)$ to itself.
4. Deduce that $\overline{K}(\mathbb{F}_8)$ contains 21 points P_{ij} in addition to the $Q_\ell : P_{ij} = \tau^i(\psi^j(P_{00}))$, where $P_{00} = (1, \alpha^2, \alpha^2 + \alpha) \in K(\mathbb{F}_8)$

Question 5. Let $Y \subset \mathbb{A}^n/k$ be an affine variety and define $A(Y)$ by the quotient

$$\pi : k[x_1, \cdots, x_n] \to A(Y) := k[x_1, \cdots, x_n]/I(Y)$$

.

1. Show that $V_Y(J) = V(\pi^{-1}(J))$ for every $J \trianglelefteq A(Y)$.
2. SHow that $\pi^{-1}(I_Y(X)) = I(X)$ for every affine subvariety $X \subset Y$.
3. Using the fact that $I(V(J)) \subset \sqrt{J}$ for every $J \trianglelefteq k[x_1, \cdots, x_n]$, deduce that $I_Y(V_Y(J)) \subset \sqrt{J}$ for every $J \trianglelefteq A(Y)$

Conclude that there is an inclusion-reversing bijection

$$\{\text{Affine subvarieties of } Y\} \Longleftrightarrow \{\text{Radical ideals in } A(Y)\}.$$

Question 6. Let k be a positive integer and let $d_k(n)$ be the coefficients satisfying

$$\zeta^k(s) = \sum_{n \geq 1} \frac{d_k(n)}{n^s}, \qquad \text{for } \operatorname{Re}(s) > 1.$$

1. Use the Euler product definition of $\zeta(s)$ for $\operatorname{Re}(s) > 1$, together with the Taylor series expansion

$$(1 - s)^{-k} = \sum_{\ell=0}^{\infty} \frac{\Gamma(k + \ell)}{\Gamma(k)\, \ell!} s^\ell, \qquad \text{valid for } \operatorname{Re}(s) < 1,$$

   to show that

$$d_k(p^\ell) = \frac{\Gamma(k + \ell)}{\Gamma(k)\, \ell!}.$$

   for prime powers p^ℓ. By multiplicativity, we can extend to get a formula for $d_k(n) \forall n$.

2. Prove that $d_k(n)$ counts the number of ways to write an integer n as a product of k integers. Notice that this interpretation is only valid for k an integer, but the formula in the previous point is valid for all $k > 0$.

3. Prove that $d_k(p) = k$ and that $d_k(p^\ell) \ll p^{\ell\varepsilon}$ for $\ell \geq 2$ for any $\varepsilon > 0$

4. Prove that for $X \ll T^{1-\varepsilon}(\varepsilon > 0\,\text{small})$, we have

$$\int_T^{2T} \left| \sum_{n \leq X} \frac{d_k(n)}{n^{\frac{1}{2+it}}} \right|^2 dt \sim T \sum_{n \leq X} \frac{(d_k(n))^2}{n} \ll T \prod_{p \leq X} \left(1 + \frac{|d_k(p)|^2}{p} + \frac{|d_k(p^2)|^2}{p} + \cdots \right),$$

and conclude by (3) that the Euler product behaves more or less like

$$\prod_{p \leq X} \left(1 + \frac{k^2}{p} \right).$$

5. Use $1 + x \leq e^x$ to conclude that

$$\int_T^{2T} \left| \sum_{n \leq X} \frac{d_k(n)}{n^{\frac{1}{2+it}}} \right|^2 dt \ll T(\log X)^{k^2}.$$

Question 7. We assume a regular category $\mathcal{C}$.

- Show that an arrow $g : X \to Y$ is a regular epi precisely if the following condition holds: for every commutative diagram

$$\begin{tikzcd} X \arrow[r, "b"] \arrow[d, "g"'] & A \arrow[d, "m"] \\ Y \arrow[r, "a"] & B \end{tikzcd}$$

with m mono, there is a unique arrow $h : Y \to A$ such that $mh = a$ and $hg = b$.

- Use part (a) to show that for any composable pair $A \xrightarrow{g} B \xrightarrow{f} C$ of arrows of $\mathcal{C}$ we have: if fg is regular epi, then so is f.

- For any arrow $f : X \to Y$ in $\mathcal{C}$, we define the *graph* of f as the subobject of $X \times Y$ represented by the mono $\langle \mathrm{id}_X, f \rangle : X \to X \times Y$.
Suppose X, Y, Z are objects of $\mathcal{C}$, $g : Z \to Y$ is a regular epi, and $R \in \mathrm{Sub}(X \times Y)$. Let $S = (\mathrm{id}_X \times g)^*(R) \in \mathrm{Sub}(X \times Z)$. Assume that the following two sequents of regular logic are true, with the evident interpretation:

$$\vdash_x \ \exists z\, S(x, z),$$

$$S(x, z) \wedge S(x, z') \vdash_{x,z,z'} z = z'.$$

Prove that there exists an arrow $f : X \to Y$ such that R is the graph of f.

Question 8. Let $\mathbf{A}$ be a small cartesian closed category. Prove that the *Yoneda embedding*

$$\mathbf{A} \to [\mathbf{A}^{op}, \mathbf{Set}]$$

preserves the entire cartesian closed structure (exponentials as well as products).

Question 9. Let k be a number field and let $S \subset M_k$ be a finite set containing all archimedean places. Let Λ be an infinite index set and $\mathcal{H} := \{H_1, \ldots, H_q\}$ be a set of moving hyperplanes in $\mathbb{P}^M(k)$ indexed by Λ.

Let

$$\mathbf{x} = [x_0 : \cdots : x_M] : \Lambda \to \mathbb{P}^M(k)$$

be a sequence of points. Assume that

(i) $\mathbf{x}$ is linearly nondegenerate with respect to $\mathcal{H}$, which means that for each infinite coherent subset $A \subset \Lambda$ with respect to $\mathcal{H}$, $x_0|_A, \ldots, x_M|_A$ are linearly independent over $\mathcal{R}_{A,\mathcal{H}}$;

(ii) $h(H_j(\alpha)) = o(h(\mathbf{x}(\alpha)))$ for all $\alpha \in \Lambda$ and $j = 1, \ldots, q$.

Then, for any $\epsilon > 0$, there exists an infinite index subset $A \subset \Lambda$ such that

$$\log \prod_{v \in S} \max_K \prod_{j \in K} \left(\frac{\|\mathbf{x}(\alpha)\|_v \, \|H_j(\alpha)\|_v}{\|H_j(\alpha)(\mathbf{x}(\alpha))\|_v} \right) \leq (M + 1 + \epsilon) h(\mathbf{x}(\alpha))$$

holds for all $\alpha \in A$.

Here the maximum is taken over all subsets $K \subset \{1, \ldots, q\}$ with $\#K = M + 1$ such that $H_j(\alpha)$, $j \in K$, are linearly independent over k for each $\alpha \in \Lambda$.

Question 10. Consider a *d-dimensional quantum* system with *Hamiltonian* H, with eigenvalue decomposition $H = \sum_{i=1}^{d} \epsilon_i \, |e_i\rangle\langle e_i|$. A state of that system is passive if and only if $[\rho, H] = 0$ and, for eigenvalue decomposition $\rho = \sum_{i=1}^{d} p_i \, |e_i\rangle\langle e_i|$, we have that $\epsilon_i \leq \epsilon_j$ implies $p_i \geq p_j$ for all i and j in $\{1, \cdots, d\}$.

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This is the end of this sample question paper for the examination **M.U.E.E.**, **2026**. All of the above 10 questions have been selected and designed by **Prof**. (Dr.) **A**. **S**.

MU wishes you Excellence & Success.

Prof. (Dr.) **A**. **S**.
Founder & Director
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